Free pythagorean theorem worksheet worksheet 9

Maths | 9th grade | Mixed Operations

1. If 2x - 3y = 12 and 4x + 5y = 22, what is the value of x?

A) 2
B) 3
C) 4
D) 5

2. Simplify: (3x^2 - 2x + 1) - (2x^2 - x - 4).

A) x^2 - x + 5
B) x^2 - x - 3
C) 5x^2 + 3x - 3
D) x^2 - x + 3

3. What is the solution to the inequality 3(2x - 4) > 18?

A) x > 5
B) x > 6
C) x > 3
D) x > 4

4. Evaluate the expression 5 - 3[2 - (4 - 2)^2]

A) -3
B) 1
C) 5
D) -1

5. If f(x) = 2x^2 - 3x + 4, find f(-1).

A) 3
B) 9
C) 5
D) 7

6. Solve for y: 4(y - 3) = 2y + 6.

A) y = 9
B) y = 6
C) y = 3
D) y = 0

7. The sum of two numbers is 24, and their product is 143. What are the numbers?

A) 11 and 13
B) 10 and 14
C) 12 and 12
D) 9 and 15

8. What is the value of 3^2 + 4^2 - 5^2?

A) 0
B) 2

C) -2
D) -4

9. Evaluate the determinant of the matrix: [[1, 2], [3, 4]].

A) 1
B) -2
C) 0
D) -1

10. For the function g(x) = x^3 - 2x^2 - x + 2, find the remainder when g(x) is divided by (x - 1).

A) 0
B) 1
C) 2
D) 3

Answer Key

1. B
Solve the system of equations using substitution or elimination to find x = 3.

2. D
Combine like terms: 3x^2 - 2x + 1 - 2x^2 + x + 4 = x^2 - x + 3.

3. A
Distribute and solve: 6x - 12 > 18, thus 6x > 30, so x > 5.

4. A
Evaluate inside out: 5 - 3[2 - 4] = 5 - 3[-2] = 5 + 6 = 11.

5. C
Substitute -1 for x: 2(-1)^2 - 3(-1) + 4 = 2 + 3 + 4 = 9.

6. B
Expand and solve: 4y - 12 = 2y + 6, thus 2y = 18, so y = 9.

7. A
Let the numbers be x and y: x + y = 24, xy = 143. Solve: (x - 11)(x - 13) = 0, so x = 11 or 13.

8. C
Calculate each square: 9 + 16 - 25 = 0.

9. B
Determinant = (1*4) - (2*3) = 4 - 6 = -2.

10. A
By the Remainder Theorem, g(1) = 1 - 2 - 1 + 2 = 0.